

FH Aachen

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Expose

Evaluating AI Agent Capabilities in SysML v2 Modeling via a Semantic MCP Interface

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1 Expose

Model-Based Systems Engineering (MBSE) manages the growing complexity of Cyber-Physical Systems through a central, formalized model [1]. SysML has become widely adopted as the standard for modeling such systems and serves as a key enabler of MBSE [1]. It allows discipline specific subsystems to be modeled within a shared model, and its most recent revision, SysML v2, enhances and simplifies this capability through a formal textual syntax and a standardized API [2].

This thesis investigates the modeling capabilities of LLM-based agents [3] operating interactively on a SysML v2 model through the industrial modeling tool Magic Systems of Systems Architect (MSoSA), operating on the open-source Airbus Apollo 11 SysML v2 model [4] as system under test. A purpose-built Model Context Protocol (MCP) bridge acts as a harness that allows the agent to interact with and modify the live model, with built-in guardrails that improve LLM adherence.

A recent review finds interactive, tool using agents on live models entirely absent from the literature [5]. The closest work evaluates file-based tooling only [6] or contributes an MCP framework with no capability evaluation [7], and no standard benchmark for fundamental modeling tasks exists [8]. Closing this gap matters, as industry deploys agentic MBSE tooling faster than it is measured.

RQ: *Which SysML v2 modeling tasks can an LLM-based agent perform reliably in an industrial modeling environment (MSoSA) when connected through an MCP bridge, and where do its capabilities break down?*

We hypothesize that bare MCP tool access, without an accompanying knowledge base, guardrails, and encoded best practices, produces a failure rate high enough to be counterproductive for engineering work, while a well-structured harness measurably improves output quality. However, given the non-deterministic nature of LLMs, reducing this failure rate to a negligible level is expected to require substantial iterative refinement of the harness rather than a single design pass.

The MCP bridge attaches to MSoSA's Java OpenAPI plugin and exposes a broad range of tools to the live model state [9]. These provide the interaction surface needed to exercise the following benchmark use-cases: **Create**, **Query**, **Patch**, **Delete**, **Validation** [10, 11], and **Verification** [10]. These primitives are composed across multiple tiers of difficulty into increasingly complex, realistic modeling flows, such as fault correction [12]. Each use case is benchmarked on Apollo 11 across multiple difficulty tiers on Claude-family models, allowing for both a qualitative and a quantitative assessment of the current capabilities.

Bibliography

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- [2] OBJECT MANAGEMENT GROUP: Kernel Modeling Language (KerML), Version 1.0 / Object Management Group. Version: März 2026. <https://www.omg.org/spec/KerML/1.0/PDF>. 2026 (formal/26-03-01). – Specification. – Normative OMG specification of the logic-grounded kernel SysML v2 extends, with four declarable tool conformance levels. Supplies the vocabulary for what an MCP bridge exposes – structural CRUD versus semantic conformance – and why v2 constraints are natively verifiable.
- [3] DEHART, John K.: Leveraging Large Language Models for Direct Interaction with SysML v2. In: *INCOSE International Symposium* 34 (2024), Juli, No. 1, p. 2168–2185. <http://dx.doi.org/10.1002/iis2.13262>. – DOI 10.1002/iis2.13262. – Founding paper of the LLM×SysML v2 topic: conceptual analysis and case studies, no controlled evaluation or ground truth. Provides the vision the thesis tests empirically and the “LLM edits text directly” pattern behind the no-tool arm.
- [4] HELLE, Philipp ; SCHRAMM, Gerrit: Fly me to the Moon – Modeling Apollo 11 Using SysML v2. In: *Systems Engineering* (2026), Juni. <http://dx.doi.org/10.1002/sys.70074>. – DOI 10.1002/sys.70074. – [TOP 3] Airbus Central R&T’s five-layer CoSMA (Purpose/Operational/Functional/Logical/Technical) SysML v2 model of the Apollo 11 mission, released open-source (github.com/airbus/apollo-11-sysml-v2, MPL-2.0, commit 6e9c93f, accessed 20.09.2026) as a public MBSE benchmark. The instrument this thesis measures on: ~7.2 kLOC / 28 files / ~2 000 elements, deliberately incomplete “scaffold” for *Erstellen/Korrektur* tasks.
- [5] FRESEMANN, Carina ; ELLSEL, Claudius ; GRUNENWALD, Carl-Philipp ; STARK, Rainer: Using Large Language Models for System Modeling: Current State and Future Implications. In: *2025 IEEE International Symposium on Systems Engineering (ISSE)*, IEEE, Oktober 2025, p. 1–8. – Structured review of 20 publications on LLMs for MBSE tasks (creation, analysis, modification, reformulation). Supplies the taxonomy behind this thesis’s use-case catalogue, names MCP as the emerging context mechanism, and flags the missing evaluation criteria.

- [6] DEMPSEY, Colin: *Native AI Integration for Model-Based Systems Engineering: Three Layers that Make It Work*. SEI Blog, Carnegie Mellon University, <https://www.sei.cmu.edu/blog/native-ai-integration-for-model-based-systems-engineering-three-layers-that-make-it-work/>, September 2026. – [TOP 3] Controlled three-arm SysML v2 experiment (no tool / validation loop / plus MCP knowledge and skills): both tool arms reach zero syntax errors, but only full tooling lifts the pattern score (71.7 → 94.1). Direct template for this thesis’s ablation and evidence for H2.
- [7] BAZZAL, Mahmoud ; SCHLEIFER, Simon ; LUNGU, Adriana ; KRUSE, Benjamin ; GOETZ, Stefan ; KOLDEWEY, Christian ; TISSEN, Denis ; WARTZACK, Sandro J. ; DUMITRESCU, Roman: Model Context Protocol for MBSE: A Unified Interoperable Agentic Framework for Systems Modelling. In: *Proceedings of the Design Society 6* (2026), Juli, p. 2721–2730. <http://dx.doi.org/10.1017/pds.2026.10630>. – DOI 10.1017/pds.2026.10630. – [TOP 3] MCP servers on the standard SysML v2 API with creation, modification and analysis tools, shown only through two qualitative case studies without metrics. Closest competing work: it delivers the artifact class, the thesis the missing capability evaluation inside MSoSA.
- [8] BOUAMRA, Yasmine ; YUN, Bruno ; POISSON, Alexandre ; ARMETTA, Frédéric: SysTemp: A Multi-Agent System for Template-Based Generation of SysML v2. In: *Advances in Practical Applications of Agents, Multi-Agent Systems, and Computational Social Science: The PAAMS Collection – 23rd International Conference, PAAMS 2025, Lille, France* Bd. 16031, Springer Nature Switzerland, Juni 2025 (Lecture Notes in Computer Science (LNAI)), p. 41–52. – Multi-agent template pipeline for NL-to-SysML v2 generation; scaffolding reaches zero syntax errors in 4/5 toy scenarios vs. 1/5 without, syntax only. The gap citation for this thesis’s benchmark (“absence of standard benchmarks”) and for SysML v2 corpus scarcity.
- [9] AHLBRECHT, Alexander ; ZAESKE, Wanja ; DURAK, Umut: MBSqlE: Enabling SQL as Powerful Query Language for SysML v2 in Aviation. In: *44th AIAA DATC/IEEE Digital Avionics Systems Conference (DASC 2025)*. Montreal, Canada : IEEE, September 2025. – The standard SysML v2 API offers no graph traversal (REST traversals >10 s); MBSqlE imports models into SQLite so recursive SQL traverses them in one query. Motivates a semantic/cache layer instead of a thin REST proxy for *Abfragen*.
- [10] ISO/IEC/IEEE: Systems and software engineering – System life cycle processes / International Organization for Standardization. Version: Mai 2023. <https://www.iso.org/standard/81702.html>. 2023 (ISO/IEC/IEEE 15288:2023). – International Standard. – Defines Verification (conformance to specified requirements) and Validation (fulfilment of stakeholder needs in the operational environment) as distinct life-cycle

processes. The thesis adopts these definitions for its *Verifizierung* and *Validierung* use cases and audits prior work against them.

- [11] CIBRIÁN, Eduardo ; OLIVERT-INSERTE, Jose ; DÍEZ-FENOY, Carlos ; MENDIETA, Roy ; LLORENS, Juan ; ÁLVAREZ-RODRÍGUEZ, Jose M.: Ensuring Semantic Consistency in SysML v2 Models Through Metamodel-Driven Validation. In: *IEEE Access* 13 (2025), p. 121444–121457. <http://dx.doi.org/10.1109/ACCESS.2025.3587786>. – DOI 10.1109/ACCESS.2025.3587786. – Metamodel-derived rule checking (lexical, ANTLR syntax, semantic cross-references) detects 97 % of 428 injected errors across 25 OMG test models. Precedent for *Verifizierung*: an error taxonomy, a fault-injection method transferable to *Korrektur*, and a detection-rate baseline.
- [12] AL-SHAMI, Haitham ; MALIK, Rohail ; ALA-LAURINAHO, Riku ; VEPSÄLÄINEN, Jari ; VIITALA, Raine: Automated Semantic Fault Localization in SysML v2: A Human-in-the-Loop Framework Using Knowledge-Graph Augmented LLMs. In: *INCOSE International Symposium* 36 (2026), Juli, No. 1, p. 1274–1286. <http://dx.doi.org/10.1002/iis2.70189>. – DOI 10.1002/iis2.70189. – Fine-tuned small LLM plus a physical-compatibility KG repairs *semantic* faults that pass SysML v2 syntax checks, rising from <3 % to >91 % on 1 184 test samples. Methodological precedent for the fault injection and fault taxonomy of *Korrektur*.
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- [14] GESELLSCHAFT FÜR SYSTEMS ENGINEERING (GFSE): *SysML-v2-Models: An Open Collection of SysML v2 Models*. GitHub repository, <https://github.com/GfSE/SysML-v2-Models>, commit ebbb0c3, accessed 20.09.2026, 2025. – BSD-3. ~5.1 kLOC curated teaching/reference models (VehicleModel, DroneModelLogical, EveOnline Mining Frigate) explicitly meant to "help train LLMs"; ships `CI/parse.py` + a pilot-implementation jar as a validity oracle. Secondary/generalisation set (D2) – H3 target, small self-contained models plus deliberate "bad examples".
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tober 2025, p. 1–7. – Four-component structured prompting for NL→SysML v2 generation, evaluated on an automotive use case with precision/recall/F1 and syntax checks over three runs; few-shot examples prove essential. Primary reference for *Erstellen*, but file-based without a modelling tool.

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adopted principles: external evaluator, tasks grounded in an existing model, and the repair/generation/tuning split.

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